

Lemma: A 是有限集, $B \subseteq A$, $|B| = |A|$. 则有: $A = B$.

Proof: 定义映射 $\iota: B \rightarrow A$, 显然 $\iota: B \rightarrow A$ 是一个单射
 $x \mapsto x$

$\because A$ 是有限集, $B \subseteq A \quad \therefore B$ 是有限集.

$\therefore A$ 和 B 是两个等势的有限集, $\iota: B \rightarrow A$ 是单射 $\therefore \iota: B \rightarrow A$ 是双射.

$\therefore$ 对 $\forall \alpha \in A$, $\exists \beta \in B$, s.t. $\iota(\beta) = \alpha \quad \therefore \alpha = \iota(\beta) = \beta \in B$

$\therefore A \subseteq B \quad \therefore A = B. \quad \square$

Lemma: B 是有限集, $B \subseteq A$, $|B| = |A|$. 则有: $A = B$

Proof: $\because B$ 是有限集 $\therefore |B| \in \mathbb{Z}_{\geq 0} \quad \because |A| = |B| \quad \therefore |A| \in \mathbb{Z}_{\geq 0}$.

$\therefore A$ 是有限集. $\therefore A = B. \quad \square$

Lemma: G 是群, $H \subseteq G$, H 是 G 的有限子集. 对 $\forall x \in G$, 下列

命题等价:

$$\textcircled{1} \quad xHx^{-1} \subseteq H$$

$$\textcircled{2} \quad H \subseteq xHx^{-1}$$

$$\textcircled{3} \quad xHx^{-1} = H \iff x \in N_G(H)$$

Proof: $\textcircled{1} \Rightarrow \textcircled{3}$
定义映射 $f: H \rightarrow xHx^{-1}$
$$h \mapsto xhx^{-1}$$

对 $\forall h \in H$, 有: $f(h) = xhx^{-1} \in xHx^{-1} \quad \therefore f(H) \subseteq xHx^{-1}$

显然相等的自变量给出相等的映射值

对 $\forall h_1, h_2 \in H$, 若 $f(h_1) = f(h_2)$, 则 $xh_1x^{-1} = xh_2x^{-1} \quad \therefore h_1 = h_2$

$\therefore f$ 是单射. 显然 f 是满射 $\therefore f$ 是双射. $\therefore |H| = |xHx^{-1}|$

$\therefore H$ 是有限集, $xHx^{-1} \subseteq H$, $|xHx^{-1}| = |H| \quad \therefore xHx^{-1} = H$.

$\textcircled{1} \Rightarrow \textcircled{3}$ 得证. $\textcircled{3} \Rightarrow \textcircled{1}$ 显然. $\therefore \textcircled{1} \iff \textcircled{3}$.

$\textcircled{2} \Rightarrow \textcircled{3}$. 定义映射 $f: H \rightarrow xHx^{-1}$ 已经证明了 f 是双射.
$$h \mapsto xhx^{-1}$$

$$\therefore |H| = |xHx^{-1}|$$

$\therefore H$ 是有限集, $H \subseteq xHx^{-1}$, $|H| = |xHx^{-1}| \quad \therefore xHx^{-1} = H$.

$\textcircled{2} \Rightarrow \textcircled{3}$ 得证. $\textcircled{3} \Rightarrow \textcircled{2}$ 显然. $\therefore \textcircled{2} \iff \textcircled{3}$.



Lemma: G 是群, $g \in G$, $E \subseteq G$, 定义集合 $gE = \{gx \mid x \in E\}$.

则有:

$$\textcircled{1} gE \subseteq G$$

$$\textcircled{2} |gE| = |E|$$

Proof: 若 $E = \emptyset$, 则 $gE = \{gx \mid x \in E\} = \emptyset$. $\therefore gE = \emptyset \subseteq \overset{G}{\cancel{G}}$

且 $|gE| = |E| = 0$. 下设 $E \neq \emptyset$.

任取 $gx \in gE$ (其中 $x \in E$). $\because x \in E \therefore x \in G \therefore gx \in G$.

$$\therefore gE \subseteq G.$$

定义映射 $f: E \rightarrow gE$. 显然 $f(E) \subseteq gE$ 且 f 是映射
 $x \mapsto gx$.

对 $\forall x_1, x_2 \in E$, 若 $f(x_1) = f(x_2)$, 则 $gx_1 = gx_2 \therefore g^{-1}gx_1 = g^{-1}gx_2$

$\therefore x_1 = x_2 \therefore f$ 是单射. 显然 f 是满射. $\therefore f$ 是双射.

$$\therefore |gE| = |E|.$$

Lemma: G 是群, $g_1, g_2 \in G$, $E \subseteq G$, 则有:

$$(g_1 g_2)E = g_1(g_2 E).$$

Proof: $\because E \subseteq G, g_2 \in G \therefore g_2 E \subseteq G. \therefore g_1 \in G$

$$\because g_1(g_2 E) \subseteq G. \quad \because g_1 g_2 \in G \quad \therefore (g_1 g_2) E \subseteq G$$

任取 $(g_1 g_2) E$ 中的一元: $(g_1 g_2) x \quad (x \in E).$

$$\text{有: } g_2 x \in g_2 E. \quad g_1(g_2 x) \in g_1(g_2 E)$$

$$\therefore (g_1 g_2) x \in g_1(g_2 E) \quad \therefore (g_1 g_2) E \subseteq g_1(g_2 E)$$

任取 $g_1(g_2 E)$ 中的一元: $g_1 \alpha \quad (\alpha \in g_2 E).$

$$\because \alpha \in g_2 E \quad \therefore \exists \beta \in E, \text{ s.t. } \alpha = g_2 \beta$$

$$\therefore g_1 \alpha = g_1(g_2 \beta) = (g_1 g_2) \beta \in (g_1 g_2) E.$$

$$\therefore g_1(g_2 E) \subseteq (g_1 g_2) E \quad \therefore (g_1 g_2) E = g_1(g_2 E). \quad \square$$

Lemma: G 是群, $g \in G, E \subseteq G$, 定义集合 $E_g = \{xg \mid x \in E\}$

则有:

$$\textcircled{1} E_g \subseteq G$$

$$\textcircled{2} |E_g| = |E|.$$

Proof: $\textcircled{1}$ 显然. $f: E \rightarrow E_g$ 是双射. $\therefore |E_g| = |E|$

$$x \mapsto xg$$

$\square$

Lemma: G 是 p -群, 则 G 的子群和商群都是 p -群

Proof: $|G| = p^m, m \in \mathbb{Z}_{\geq 0}$ $H \leq G \quad \therefore |H| \mid |G|$

$\therefore |H| = p^k \quad k \in \mathbb{Z}_{\geq 0} \text{ 且 } k \leq m \quad \therefore H \text{ 是 } p\text{-群}.$

$\therefore p$ -群的子群是 p -群.

设 $H \triangleleft G$. $\therefore G/H$ 是群. 设 $|H| = p^k, k \in \mathbb{Z}_{\geq 0} \text{ 且 } k \leq m.$

$$\therefore |G/H| = \frac{|G|}{|H|} = \frac{p^m}{p^k} = p^{m-k} \quad (m-k \in \mathbb{Z}_{\geq 0} \text{ 且 } m-k \leq m)$$

$\therefore G/H$ 是 p -群.

$\therefore p$ -群的商群是 p -群. $\square$

Lemma: G 是群, $H \leq G, g \in G$. 已经证明了 $gHg^{-1} \leq G$.

称 gHg^{-1} 为 H 对 g 的共轭子群. 则有: $|gHg^{-1}| = |H|$.

Proof: 定义映射 $f: H \rightarrow gHg^{-1}$ 显然 f 是映射.

$$x \mapsto gxg^{-1}$$

又对 $\forall x_1, x_2 \in H$, 若 $f(x_1) = f(x_2)$, 则 $gx_1g^{-1} = gx_2g^{-1}$

$\therefore x_1 = x_2 \quad \therefore f$ 是单射. 显然 f 是满射. $\therefore f$ 是双射.

$$\therefore |H| = |gHg^{-1}|. \quad \square$$

Lemma: G 是群, $H \leq G$, $G/H = \{gH \mid g \in G\} = \{xH \mid x \in G\}$

定义映射 $\alpha: G \times G/H \rightarrow G/H$

$$(g, xH) \mapsto (gx)H.$$

则映射 α 是群 G 在 G/H 上的群作用.

Proof: 对 $\forall (g, xH) \in G \times G/H$, 有: $g \in G$ 且 $xH \in G/H$

(其中 $x \in G$). $\because g \in G$ 且 $x \in G \therefore gx \in G \therefore (gx)H \in G/H$

$$\therefore \alpha(g, xH) = (gx)H \in G/H$$

对 $\forall (g_1, x_1H), (g_2, x_2H) \in G \times G/H$, 若

$(g_1, x_1H) = (g_2, x_2H)$, 则 $g_1 = g_2$ 且 $x_1H = x_2H$ ~~$(g_1, x_1H) = (g_2, x_2H)$~~

$\therefore g_1 = g_2 \in G$ 且 $x_1, x_2 \in G$, $x_1^{-1}x_2 \in H$.

$\therefore$ ~~$g_1x_1 \in G$~~ $g_1x_1 \in G$ 且 $g_2x_2 \in G$,

$$(g_1x_1)^{-1}(g_2x_2) = x_1^{-1}g_1^{-1}g_2x_2 = x_1^{-1}x_2 \in H$$

$$\therefore (g_1x_1)H = (g_2x_2)H$$

$$\therefore \alpha(g_1, x_1H) = (g_1x_1)H = (g_2x_2)H = \alpha(g_2, x_2H)$$

$\therefore \alpha$ 是良定义的映射.

对 $\forall xH \in G/H$ (其中 $x \in G$), 有: $\alpha(1_G, xH) = (1_G x)H$
 $= xH$

对 $\forall g_1, g_2 \in G, \forall xH \in G/H$ (其中 $x \in G$), 有:

$$\alpha(g_1, \alpha(g_2, xH)) = \alpha(g_1, (g_2 x)H) = (g_1 (g_2 x))H \\ = ((g_1 g_2) x)H = \alpha(g_1 g_2, xH)$$

$\therefore \alpha$ 是群 G 在 G/H 上的群作用. $\square$

定义 (G -集) G 是一个群, X 是非空集合, 存在群 G 在集合 X 上的群作用, 则称 X 是一个 G -集.

Lemma: p 是素数, $a, b, m \in \mathbb{Z}_{\geq 0}$, $b \leq a$, $a \neq 0$, 则有:

$$\binom{p^m a}{p^m b} \equiv \binom{a}{b} \pmod{p}$$

Proof: $\because p$ 是素数 $\therefore$ 已经证明了 $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$ 是域, $\mathbb{F}_p[X]$ 是 Euclid 环, $\mathbb{F}_p[X]$ 是主理想环, $\mathbb{F}_p[X]$ 是唯一分解环, $\mathbb{F}_p[X]$ 是交换环.

$\because p$ 是素数 $\therefore$ 在环 $\mathbb{F}_p[X]$ 中, 对于 $m \in \mathbb{Z}_{\geq 0}$, 有:

$$\left(X + 1_{\mathbb{F}_p[X]}\right)^{p^m} = X^{p^m} + 1_{\mathbb{F}_p[X]}$$

~~$\because$~~ $\because a \in \mathbb{Z}_{\geq 0}$ 且 $a \neq 0 \therefore a \in \mathbb{Z}_{\geq 1}$

$$\therefore \left(X + 1_{\mathbb{F}_p[X]}\right)^{p^m a} = \left(\left(X + 1_{\mathbb{F}_p[X]}\right)^{p^m}\right)^a = \left(X^{p^m} + 1_{\mathbb{F}_p[X]}\right)^a$$

$\because \mathbb{F}_p[X]$ 是交换环, $X \in \mathbb{F}_p[X]$, $1_{\mathbb{F}_p[X]} \in \mathbb{F}_p[X]$, $p^m a \in \mathbb{Z}_{\geq 1}$

$$\therefore \left(X + 1_{\mathbb{F}_p[X]}\right)^{p^m a} = \sum_{k=0}^{p^m a} \binom{p^m a}{k} X^k \cdot 1_{\mathbb{F}_p[X]}^{p^m a - k} = \sum_{k=0}^{p^m a} \binom{p^m a}{k} X^k$$

$\because b \leq a, p^m > 0 \therefore p^m b \leq p^m a \therefore p^m b \in \mathbb{Z}_{\geq 0}$

$\therefore p^m b \in \mathbb{Z}_{\geq 0}$ 且 $p^m b \leq p^m a$

~~$\therefore (X + 1)$~~

$\therefore$ 在 $(X + 1_{\mathbb{F}_p[X]})^{p^m \cdot a}$ 的展开式中, $X^{p^m \cdot b}$ 这一项的系数为 $\binom{p^m \cdot a}{p^m \cdot b}$

$\therefore \mathbb{F}_p[X]$ 是交换环, $X^{p^m} \in \mathbb{F}_p[X]$, $1_{\mathbb{F}_p[X]} \in \mathbb{F}_p[X]$, $a \in \mathbb{Z}_{\geq 1}$

$$\therefore (X^{p^m} + 1_{\mathbb{F}_p[X]})^a = \sum_{k=0}^a \binom{a}{k} (X^{p^m})^k \cdot 1_{\mathbb{F}_p[X]}^{a-k} = \sum_{k=0}^a \binom{a}{k} X^{p^m \cdot k}$$

$\therefore b \in \mathbb{Z}_{\geq 0}$ 且 $b \leq a$

$\therefore$ 在 $(X^{p^m} + 1_{\mathbb{F}_p[X]})^a$ 的展开式中, $X^{p^m \cdot b}$ 这一项的系数为 $\binom{a}{b}$

$$\therefore (X + 1_{\mathbb{F}_p[X]})^{p^m \cdot a} = (X^{p^m} + 1_{\mathbb{F}_p[X]})^a \in \mathbb{F}_p[X]$$

$$\therefore \text{在 } \mathbb{F}_p \text{ 中有 } \binom{p^m \cdot a}{p^m \cdot b} = \binom{a}{b} \quad \therefore \binom{p^m \cdot a}{p^m \cdot b} \equiv \binom{a}{b} \pmod{p}$$

$$\begin{aligned} \text{Remark: 在 } \mathbb{F}_p[X] \text{ 中, } \binom{p^m \cdot a}{p^m \cdot b} X^{p^m \cdot b} &= \left(\binom{p^m \cdot a}{p^m \cdot b} \cdot 1_{\mathbb{F}_p[X]} \right) \cdot X^{p^m \cdot b} \\ &= \left(\binom{p^m \cdot a}{p^m \cdot b} \cdot [1]_p \right) \cdot X^{p^m \cdot b} = \left[\binom{p^m \cdot a}{p^m \cdot b} \right]_p \cdot X^{p^m \cdot b} \end{aligned}$$

$$\begin{aligned} \text{在 } \mathbb{F}_p[X] \text{ 中, } \binom{a}{b} X^{p^m \cdot b} &= \left(\binom{a}{b} \cdot 1_{\mathbb{F}_p[X]} \right) \cdot X^{p^m \cdot b} = \left(\binom{a}{b} \cdot [1]_p \right) \cdot X^{p^m \cdot b} \\ &= \left[\binom{a}{b} \right]_p X^{p^m \cdot b} \end{aligned}$$

$$\therefore \left[\binom{p^m \cdot a}{p^m \cdot b} \right]_p = \left[\binom{a}{b} \right]_p \quad \therefore \binom{p^m \cdot a}{p^m \cdot b} \equiv \binom{a}{b} \pmod{p} \quad \square$$

Lemma: G 是有限群, p 是素数, $H \leq G$, H 是 p -群, 则有:

$$\textcircled{1} (G:H) \equiv (N_G(H):H) \pmod{p}$$

$$\textcircled{2} \text{ 若 } p \mid (G:H), \text{ 则 } p \mid (N_G(H):H) \text{ 且 } H < N_G(H)$$

Proof: $\because G$ 是有限群, $H \leq G \therefore H$ 是有限群 $\therefore |G| \in \mathbb{Z}_{\geq 1}, |H| \in \mathbb{Z}_{\geq 1}$

$$\therefore |G| = |H| \cdot (G:H) \therefore (G:H) \in \mathbb{Z}_{\geq 1} \therefore |G/H| = (G:H) \in \mathbb{Z}_{\geq 1}$$

$\because G$ 是有限群, $H \leq G \therefore N_G(H) \leq G \therefore N_G(H)$ 是有限群

$$\therefore |N_G(H)| \in \mathbb{Z}_{\geq 1} \therefore G \text{ 是有限群, } H \leq G \therefore H < N_G(H) \therefore H \leq N_G(H)$$

$$\therefore |N_G(H)| = |H| \cdot (N_G(H):H), \therefore (N_G(H):H) \in \mathbb{Z}_{\geq 1}$$

$$\therefore G \text{ 是有限群, } H \leq G \therefore G/H = \{xH \mid x \in G\}$$

定义 H 在 G/H 上的群作用为: $H \times G/H \rightarrow G/H$
$$h \cdot (xH) \mapsto (hx)H$$

对 $\forall h \in H, \forall xH \in G/H$, 有: $\because h \in H, x \in G \therefore hx \in G \therefore (hx)H \in G/H$

$$\therefore h \cdot (xH) = (hx)H \in G/H.$$

对 $\forall h_1, h_2 \in H, \forall x_1H, x_2H \in G/H$. 若 $(h_1, x_1H) = (h_2, x_2H)$,

则有: $h_1 = h_2$ 且 $x_1H = x_2H \therefore h_1 = h_2$ 且 $x_1^{-1}x_2 \in H$

$$\therefore G \text{ 是有限群, } H \leq G, h_1x_1 \in G, h_2x_2 \in G$$

$$(h_1 x_1)^{-1} (h_2 x_2) = x_1^{-1} h_1^{-1} h_2 x_2 = x_1^{-1} h_1^{-1} h_1 x_2 = x_1^{-1} |_G x_2 = x_1^{-1} x_2 \in H$$

$$\therefore (h_1 x_1)H = (h_2 x_2)H \quad \therefore h_1 \cdot (x_1 H) = h_2 \cdot (x_2 H)$$

$$\text{对 } \forall xH \in G/H, \text{ 有: } |_G \cdot xH = (|_G \cdot x)H = xH$$

$$\text{对 } \forall h_1, h_2 \in H, \forall xH \in G/H, \text{ 有: } h_1 \cdot (h_2 \cdot xH) = h_1 \cdot (h_2 x)H \\ = (h_1 (h_2 x))H = ((h_1 h_2) x)H = (h_1 h_2) \cdot xH \quad \therefore \text{这个群作用良定义.}$$

$$\therefore H \text{ 是 } p\text{-群}, G/H \text{ 是非空有限集合} \quad \therefore |G/H| \equiv |(G/H)^H| \pmod{p}$$

$$\therefore (G:H) \equiv |(G/H)^H| \pmod{p}$$

$$\therefore (G/H)^H = \{xH \in G/H \mid \text{对 } \forall h \in H \text{ 都有 } h \cdot xH = xH\}$$

$$= \{xH \in G/H \mid \text{对 } \forall h \in H \text{ 都有 } (hx)H = xH\}$$

$$= \{xH \in G/H \mid \text{对 } \forall h \in H \text{ 都有 } x^{-1}hx \in H\}$$

$$\text{对 } \forall x \in G,$$

$$\text{若 } xH \in (G/H)^H, \text{ 则有: } xH \in G/H \text{ 且对 } \forall h \in H \text{ 都有 } x^{-1}hx \in H$$

$$\text{已知 } x \in G, H \leq G, \text{ 任取 } xH \text{ 中的元素: } x\alpha x^{-1} \text{ (其中 } \alpha \in H)$$

$$\text{对 } \forall \alpha \in H, \therefore \alpha \in H \quad \therefore x^{-1}\alpha x \in H \quad \therefore x(x^{-1}\alpha x)x^{-1} \in xHx^{-1}$$

$$\therefore \alpha \in xHx^{-1} \quad \therefore H \subseteq xHx^{-1}.$$

$$\therefore G \text{ 是有限群}, H \leq G, H \text{ 是 } G \text{ 的有限子群}, x \in G, H \subseteq xHx^{-1}$$

$$\therefore xHx^{-1} = H \quad \therefore x \in N_G(H)$$

若 $x \in N_G(H)$, 则有: $xHx^{-1} = H \quad \because x \in G \quad \therefore xH \in G/H$

$\therefore \forall h \in H$, 有: $\because h \in H = xHx^{-1} \quad \therefore \exists \beta \in H$, s.t. $h = x\beta x^{-1}$

$\therefore x^{-1}hx = x^{-1}(x\beta x^{-1})x = (x^{-1}x)\beta(x^{-1}x) = 1_G \beta 1_G = \beta \in H$

$\therefore xH \in (G/H)^H$

$\therefore \forall x \in G$, 有: $xH \in (G/H)^H \iff x \in N_G(H)$

$\because N_G(H)$ 是有限群, $H \leq N_G(H) \quad \therefore N_G(H)/H = \{xH \mid x \in N_G(H)\}$

定义映射 $\varphi: N_G(H)/H \longrightarrow (G/H)^H$

$$xH \mapsto xH$$

对 $\forall xH \in N_G(H)/H$ (其中 $x \in N_G(H)$), 有: $\because x \in N_G(H) \quad \therefore x \in G$

$\because x \in G$ 且 $x \in N_G(H) \quad \therefore xH \in (G/H)^H \quad \therefore \varphi(xH) = xH \in (G/H)^H$

显然相等的自变量映到相等的映射值, $\therefore \varphi$ 是映射.

显然 φ 是单射.

对 $\forall xH \in (G/H)^H$, 有: $xH \in G/H \quad \therefore x \in G$

$\because x \in G$ 且 $xH \in (G/H)^H \quad \therefore x \in N_G(H) \quad \therefore xH \in N_G(H)/H$.

$\because \varphi(xH) = xH \quad \therefore \varphi$ 是满射 $\therefore \varphi$ 是双射

$$\therefore |(G/H)^H| = |N_G(H)/H| = (N_G(H):H)$$

$\therefore (G:H) \equiv (N_G(H):H) \pmod{p} \quad \textcircled{1} \text{得证}.$

$$\because (G:H) \equiv (N_G(H):H) \pmod{p}, \quad p \mid (G:H)$$

$$\because p \mid (N_G(H):H) \quad \because G \text{ 是有限群}, H \leq G \quad \therefore N_G(H) \leq G$$

$$\because N_G(H) \text{ 是有限群} \quad \therefore |N_G(H)| \in \mathbb{Z}_{\geq 1} \quad \therefore H \triangleleft N_G(H)$$

$$\because H \leq N_G(H) \quad \therefore H \text{ 是有限群} \quad \therefore |H| \in \mathbb{Z}_{\geq 1}$$

$$\because |N_G(H)| = |H| \cdot (N_G(H):H) \quad \therefore (N_G(H):H) \in \mathbb{Z}_{\geq 1}$$

$$\because p \mid (N_G(H):H) \quad \therefore (N_G(H):H) = \zeta p \quad (\zeta \in \mathbb{Z}_{\geq 1})$$

$$\because (N_G(H):H) = \zeta p \geq p \geq 2 > 1 \quad \therefore |H| > 0$$

$$\because |N_G(H)| = |H| \cdot (N_G(H):H) > |H|$$

$$\because H \leq N_G(H) \text{ 且 } |H| < |N_G(H)| \quad \therefore H < N_G(H) \quad \square$$

Lemma: G 是 p -群, $H < G$, 则有: $H < N_G(H)$ 且 $H \triangleleft N_G(H)$

Proof: $\because G$ 是 p -群 $\therefore G$ 是有限群, $|G| = p^m$ ($m \in \mathbb{Z}_{\geq 0}$)

$$\because H < G \quad \therefore |H| \mid |G| \quad \therefore |H| \mid p^m \quad \because H < G \quad \therefore |H| < |G|$$

$$\therefore |H| < p^m \quad \therefore |H| = p^k \quad (k \in \mathbb{Z}_{\geq 0} \text{ 且 } k < m) \quad \therefore m \in \mathbb{Z}_{\geq 1}$$

$$\therefore H \text{ 是 } p\text{-群} \quad \because |G| = |H| \cdot (G:H) \quad \therefore (G:H) = \frac{|G|}{|H|} = \frac{p^m}{p^k} = p^{m-k} \quad (m-k \in \mathbb{Z}_{\geq 1})$$

$$\because p \mid (G:H) \quad \therefore H < N_G(H) \quad \therefore H < N_G(H) \text{ 且 } H \triangleleft N_G(H) \quad \square$$

Lemma: G 是群, $H \leq G$, $(G:H) = p$ (p 是素数) 则有:

$$H \triangleleft G \iff H \subsetneq N_G(H)$$

Proof: ($\Rightarrow$): $\because G$ 是群, $H \leq G$, $H \triangleleft G \therefore N_G(H) = G$.

~~G~~ 假设 $H = G$, 则有:

$$\because G \text{ 是群}, H \leq G \quad \therefore G/H = \{gH \mid g \in G\}.$$

$$\text{对 } \forall g \in G, \text{ 有: } g \in G = H \quad \therefore g \in H \quad \therefore gH = H \quad \therefore G/H = \{H\}$$

$$\therefore (G:H) = |G/H| = 1 < p. \text{ 矛盾. } \therefore H \neq G \quad \therefore H \subsetneq G$$

$$\therefore H \subsetneq N_G(H)$$

$$(\Leftarrow): \because G \text{ 是群}, H \leq G \quad \therefore H \triangleleft N_G(H) \quad \therefore H \leq N_G(H) \quad \therefore N_G(H) \leq G$$

$$\because G \text{ 是群}, H \leq N_G(H) \leq G \quad \therefore (G:H) = (G:N_G(H))(N_G(H):H)$$

$$\therefore (G:N_G(H))(N_G(H):H) = p \quad \therefore (G:N_G(H)) \in \mathbb{Z}_{\geq 1}, (N_G(H):H) \in \mathbb{Z}_{\geq 1}$$

$$\therefore (N_G(H):H) = 1 \text{ 或 } (N_G(H):H) = p.$$

$$\text{假设 } (N_G(H):H) = 1, \text{ 则有 } |N_G(H)/H| = 1.$$

$$\therefore \text{对 } \forall x \in N_G(H), \text{ 有: } xH = H \quad \therefore x \in H \quad \therefore N_G(H) \subseteq H \subseteq N_G(H)$$

$$\therefore H = N_G(H). \text{ 矛盾. } \therefore (N_G(H):H) = p. \quad \therefore (G:N_G(H)) = 1$$

$$\therefore ~~G~~ |G| = |N_G(H)| \cdot (G:N_G(H)) = |N_G(H)| \quad \therefore N_G(H) = G \quad \therefore H \triangleleft G$$

$\square$

定义 (Sylow p -子群) G 是 n 阶有限群 ($n \in \mathbb{Z}_{\geq 1}$), p 是素数, $m \in \mathbb{Z}_{\geq 0}$ 且 $p^m \parallel n$. 满足 $|H| = p^m$ 的子群 $H \leq G$ 称为 G 的 Sylow p -子群.

记号: G 是有限群, p 是素数, 则 G 的所有 Sylow p -子群组成的集合记作 $\text{Syl}_p(G)$.

定理 (Sylow 第一定理) G 是有限群, p 是素数, 则 G 有 Sylow p -子群.

Proof: 设 $|G| = n$. $\because G$ 是有限群 $\therefore |G| = n \in \mathbb{Z}_{\geq 1}$.

如果 $p \nmid n$, 则有 $p^0 \parallel n$ 且 $p^1 \nmid n \therefore p^0 \parallel n \therefore |\{1_G\}| = 1 = p^0$

且 $\{1_G\} \leq G \therefore \{1_G\}$ 是 G 的 Sylow p -子群.

如果 $p \mid n$, 设 $p^m \parallel n \therefore p^m \mid n$ 且 $p^{m+1} \nmid n \therefore m \in \mathbb{Z}_{\geq 1}$

定义集合 $Y = \{E \leq G \mid |E| = p^m\}$.

$\because p^m \mid n \therefore \frac{n}{p^m} \in \mathbb{Z}_{\geq 1} \therefore p^m \leq n \therefore p^m \leq |G| \therefore p^m \in \mathbb{Z}_{\geq 1}$

$\therefore G$ 中一定有元素个数为 p^m 的子集 $\therefore Y \neq \emptyset. \therefore |Y| \in \mathbb{Z}_{\geq 1}$

$\therefore |Y| = \binom{n}{p^m} \therefore p$ 是素数, $\frac{n}{p^m} \in \mathbb{Z}_{\geq 1}, 1 \in \mathbb{Z}_{\geq 1}, m \in \mathbb{Z}_{\geq 1}, 1 \leq \frac{n}{p^m}$

$$\therefore \binom{p^m \cdot \frac{n}{p^m}}{p^m, 1} \equiv \binom{\frac{n}{p^m}}{1} \pmod{p} \quad \therefore \binom{n}{p^m} \equiv \frac{n}{p^m} \pmod{p}$$

假设 $p \mid \frac{n}{p^m}$, 则 $\frac{n}{p^{m+1}} \in \mathbb{Z}_{\geq 1} \quad \therefore p^{m+1} \mid n$ 矛盾. $\therefore p \nmid \frac{n}{p^m}$

$$\therefore p \nmid \binom{n}{p^m} \quad \therefore p \nmid |Y| \quad \therefore |Y| \in \mathbb{Z}_{\geq 1} \text{ 且 } p \nmid |Y|$$

定义群 G 在集合 Y 上的群作用 $\alpha: G \times Y \longrightarrow Y$
 $(g, E) \mapsto gE$

对 $\forall (g, E) \in G \times Y$, 有: $g \in G$ 且 ~~$E \in G$~~ $E \in Y$

$$\therefore g \in G \text{ 且 } E \subseteq G \text{ 且 } |E| = p^m \quad \therefore \alpha(g, E) = gE \subseteq G$$

$$\text{且 } |\alpha(g, E)| = |gE| = |E| = p^m \quad \therefore \alpha(g, E) \in Y$$

对 $\forall (g_1, E_1), (g_2, E_2) \in G \times Y$, 若 $(g_1, E_1) = (g_2, E_2)$, 则

$$g_1 = g_2 \text{ 且 } E_1 = E_2. \quad \therefore \alpha(g_1, E_1) = g_1 E_1 = g_2 E_2 = \alpha(g_2, E_2)$$

↓
双方向包含显然.

$\therefore \alpha: G \times Y \rightarrow Y$ 是良定义的映射.

$$\text{对 } \forall E \in Y, \text{ 有: } \alpha(1_G, E) = 1_G E = \{1_G x \mid x \in E\} = \{x \mid x \in E\} = E$$

对 $\forall g_1, g_2 \in G, \forall E \in Y$, 有: $E \subseteq G$

$$\begin{aligned} \therefore \alpha(g_1, \alpha(g_2, E)) &= \alpha(g_1, g_2 E) = g_1 (g_2 E) = (g_1 g_2) E \\ &= \alpha(g_1 g_2, E) \end{aligned} \quad \therefore \alpha: G \times Y \rightarrow Y \text{ 是 } G \text{ 在 } Y \text{ 上的群作用.}$$

∴ 群 G 作用在有限非空集合 Y 上. 设所有的 G -轨道为

$$GE_1, GE_2, \dots, GE_\lambda \quad (\text{其中 } \lambda \in \mathbb{Z}_{\geq 1}, E_1, E_2, \dots, E_\lambda \in Y)$$

$$\therefore Y = \bigsqcup_{i=1}^{\lambda} GE_i \quad \text{且对 } \forall i=1, 2, \dots, \lambda, \text{ 都有 } GE_i \neq \emptyset.$$

$$\therefore |Y| = \sum_{i=1}^{\lambda} (G : \text{Stab}_G(E_i))$$

$$\text{对 } \forall i=1, 2, \dots, \lambda, (G : \text{Stab}_G(E_i)) = |G / \text{Stab}_G(E_i)| = |GE_i| \in \mathbb{Z}_{\geq 1}$$

$$\therefore p \nmid |Y| \quad \therefore \exists \eta \in \{1, 2, \dots, \lambda\}, \text{ s.t. } p \nmid (G : \text{Stab}_G(E_\eta))$$

$$\therefore E_\eta \in Y \quad \therefore \text{Stab}_G(E_\eta) \leq G. \quad \therefore |\text{Stab}_G(E_\eta)| \in \mathbb{Z}_{\geq 1}$$

$$\therefore |G| = |\text{Stab}_G(E_\eta)| \cdot (G : \text{Stab}_G(E_\eta)) \quad \text{且 } |G| \in \mathbb{Z}_{\geq 1},$$

$$|\text{Stab}_G(E_\eta)| \in \mathbb{Z}_{\geq 1}, (G : \text{Stab}_G(E_\eta)) \in \mathbb{Z}_{\geq 1}$$

$$\therefore p^m \mid |G|, \quad p \nmid (G : \text{Stab}_G(E_\eta)) \quad \therefore p^m \mid |\text{Stab}_G(E_\eta)|$$

$$\therefore p^m \leq |\text{Stab}_G(E_\eta)|$$

$$\therefore E_\eta \in Y \quad \therefore E_\eta \subseteq G \quad \text{且 } |E_\eta| = p^m \in \mathbb{Z}_{\geq 1}.$$

$$\text{任取 } d \in E_\eta. \quad \therefore d \in G \quad \text{且 } \text{Stab}_G(E_\eta) \leq G. \quad \therefore \text{Stab}_G(E_\eta) \cdot d \subseteq G$$

$$\text{任取 } \text{Stab}_G(E_\eta) \cdot d \text{ 中的一元: } \varepsilon d \quad (\text{其中 } \varepsilon \in \text{Stab}_G(E_\eta)).$$

$$\therefore \varepsilon \in \text{Stab}_G(E_\eta) \quad \therefore \varepsilon \in G \quad \text{且 } \varepsilon E_\eta = E_\eta \quad \therefore \varepsilon d \in \varepsilon E_\eta = E_\eta$$

$$\therefore \text{Stab}_G(E_1) \cdot d \subseteq E_1 \quad \because G \text{ 是群, } d \in G, \text{Stab}_G(E_1) \leq G$$

$$\therefore |\text{Stab}_G(E_1) \cdot d| = |\text{Stab}_G(E_1)|$$

$$\therefore p^m \leq |\text{Stab}_G(E_1)| = |\text{Stab}_G(E_1) \cdot d| \leq |E_1| = p^m$$

$$\therefore \text{Stab}_G(E_1) \leq G \text{ 且 } |\text{Stab}_G(E_1)| = p^m$$

$$\therefore \text{Stab}_G(E_1) \text{ 是 } G \text{ 的 Sylow } p\text{-子群.} \quad \square$$

Lemma: G 是有限群, p 是素数, P 是 G 的 Sylow p -子群, H 是 G 的 p -子群且满足 $H \subseteq N_G(P)$, 则 $H \subseteq P$ 且 $H \leq P$

Proof: $\because G$ 是有限群, $H \leq G, P \leq G, H \subseteq N_G(P)$

$\therefore$ 由群的第二同构定理得: $HP \leq G, P \triangleleft HP, H \cap P \triangleleft H$

$$H/(H \cap P) \simeq HP/P \quad \because H \text{ 是 } p\text{-群} \quad \therefore H/(H \cap P) \text{ 是 } p\text{-群}$$

$$\therefore HP/P \text{ 是 } p\text{-群.} \quad \therefore \exists k \in \mathbb{Z}_{\geq 0}, \text{ s.t. } |HP/P| = p^k$$

$$\therefore p^k = |HP/P| = (HP : P) \quad \because P \text{ 是 } G \text{ 的 Sylow } p\text{-子群}$$

$$\therefore \text{设 } |P| = p^l, l \in \mathbb{Z}_{\geq 0}.$$

$$\therefore |HP| = |P| \cdot (HP : P) = p^l \cdot p^k = p^{l+k} \quad (l+k \in \mathbb{Z}_{\geq 0})$$

$$\therefore HP \text{ 是 } p\text{-群.} \quad \because P \text{ 是 } G \text{ 的 Sylow } p\text{-子群, } |P| = p^l, l \in \mathbb{Z}_{\geq 0}$$

$$\therefore |G| = p^l \cdot \alpha, \quad \alpha \in \mathbb{Z}_{\geq 1} \text{ 且 } p \nmid \alpha$$

$$\therefore HP \leq G \quad \therefore |HP| \mid |G| \quad \therefore p^{l+k} \mid p^l \cdot \alpha \quad \therefore p^k \mid \alpha$$

$$\therefore k=0 \quad \therefore |HP/P| = p^0 = 1 \quad \therefore (HP:P) = |HP/P| = p^k = 1$$

$$\therefore |HP| = |P| \quad \therefore P \trianglelefteq HP \quad \therefore HP = P$$

$$\therefore H \subseteq HP \quad \therefore H \subseteq P \quad \therefore H \leq P \quad \square$$

定理 (Sylow 第二定理) G 是有限群, p 是素数, 则:

① ~~G 的任意~~ G 的任意 p -子群 H 皆包含于某个 G 的 Sylow p -子群

② G 的任意两个 Sylow p -子群皆共轭.

Proof: $\because G$ 是有限群, p 是素数 $\therefore G$ 有 Sylow p -子群.

设 P 是 G 的一个 Sylow p -子群. 定义集合 $X = \{gPg^{-1} \mid g \in G\}$.

记群 G 的所有子群组成的集合为 $\text{Sub}(G)$. 即 $\text{Sub}(G) = \{K \mid K \leq G\}$.

定义群作用 $\alpha: G \times \text{Sub}(G) \rightarrow \text{Sub}(G)$

$$(g, K) \mapsto gKg^{-1}$$

对 $\forall (g, K) \in G \times \text{Sub}(G)$, 有: $g \in G$ 且 $K \leq G$.

则 $\alpha(g, K) = gKg^{-1} \leq G \quad \therefore \alpha(g, K) \in \text{Sub}(G)$

对 $\forall (g_1, K_1), (g_2, K_2) \in G \times \text{Sub}(G)$, 若 $(g_1, K_1) = (g_2, K_2)$, 则

$g_1 = g_2 \in G$ 且 $K_1 = K_2 \leq G$.

$$\therefore a(g_1, K_1) = g_1 K_1 g_1^{-1} = g_2 K_2 g_2^{-1} = a(g_2, K_2)$$

双方向包含显然

$\therefore a: G \times \text{Sub}(G) \rightarrow \text{Sub}(G)$ 是良定义的映射.

$$\text{对 } \forall K \in \text{Sub}(G), \text{ 有: } K \leq G. \therefore a(g, K) = gKg^{-1} = gKg^{-1}$$

$$= \{gkg^{-1} \mid k \in K\} = \{k \mid k \in K\} = K$$

$$\text{对 } \forall g_1, g_2 \in G, \forall K \leq G, \text{ 有: } a(g_1, a(g_2, K)) =$$

$$a(g_1, g_2 K g_2^{-1}) = g_1 (g_2 K g_2^{-1}) g_1^{-1} = (g_1 g_2) K (g_2^{-1} g_1^{-1})$$

双方向包含显然

$$= (g_1 g_2) K (g_1 g_2)^{-1} = a(g_1 g_2, K)$$

$\therefore a: G \times \text{Sub}(G) \rightarrow \text{Sub}(G)$ 是群 G 在集合 $\text{Sub}(G)$ 上的群作用.

且 $\text{Sub}(G)$ 是非空有限集合.

$$\therefore P \text{ 是 } G \text{ 的一个 Sylow } p\text{-子群} \therefore P \leq G \therefore P \in \text{Sub}(G)$$

$$\therefore \text{包含 } P \text{ 的 } G\text{-轨道 } GP = \{gPg^{-1} \mid g \in G\} = X$$

$$\text{任取集合 } X \text{ 中的一个元素 } gPg^{-1} \text{ (其中 } g \in G). \therefore P \leq G, g \in G$$

$$\therefore gPg^{-1} \leq G \text{ 且 } |gPg^{-1}| = |P| \therefore gPg^{-1} \text{ 是 } G \text{ 的 Sylow } p\text{-子群.}$$

$\therefore$ 集合 X 中的任意一个元素都是 G 的 Sylow p -子群.

~~$$\therefore P \leq G \quad \text{映射 } a: G \times G/P \rightarrow G/P$$~~

~~$$(g, xP) \mapsto$$~~

$$\because P \leq G \quad \therefore N_G(P) \leq G \quad \text{映射 } \alpha: G \times N_G(P) \rightarrow G/N_G(P)$$

$\therefore$ 映射 $\alpha: G \times G/N_G(P) \rightarrow G/N_G(P)$ 是一个群作用.

$$(g, xN_G(P)) \mapsto (gx)N_G(P)$$

定义映射 $\varphi: G/N_G(P) \rightarrow X$
 $xN_G(P) \mapsto xPx^{-1}$

对 $\forall xN_G(P) \in G/N_G(P)$ (其中 $x \in G$). 有:

$$\varphi(xN_G(P)) = xPx^{-1} \in X \quad \therefore \varphi(G/N_G(P)) \subseteq X.$$

对 $\forall xN_G(P), yN_G(P) \in G/N_G(P)$ (其中 $x, y \in G$), ~~若~~
~~若~~ $xN_G(P) = yN_G(P)$, 则 $x^{-1}y \in N_G(P)$.

$$\therefore (x^{-1}y)P(x^{-1}y)^{-1} = P \quad \therefore (x^{-1}y)P(y^{-1}x) = P$$

$$\therefore xPx^{-1} = yPy^{-1}$$

(对 $\forall x\alpha x^{-1} \in xPx^{-1}$ (其中 $\alpha \in P$). $\therefore \alpha \in P \therefore \exists \beta \in P$, s.t.

$$\alpha = (x^{-1}y)\beta(y^{-1}x) \quad \therefore x\alpha x^{-1} = xx^{-1}y\beta y^{-1}xx^{-1} = y\beta y^{-1}$$

$$\therefore x\alpha x^{-1} = y\beta y^{-1} \in yPy^{-1} \quad \therefore xPx^{-1} \subseteq yPy^{-1}$$

对 $\forall y\alpha y^{-1} \in yPy^{-1}$, $\alpha \in P$, $x^{-1}y\alpha y^{-1}x \in x^{-1}yPy^{-1}x = P$

$$\therefore x(x^{-1}y\alpha y^{-1}x)x^{-1} \in xPx^{-1} \quad \therefore y\alpha y^{-1} \in xPx^{-1}$$

$$\therefore yPy^{-1} \subseteq xPx^{-1} \quad \therefore xPx^{-1} = yPy^{-1})$$

$$\therefore \varphi(xN_G(P)) = \varphi(yN_G(P)) \quad \therefore \varphi \text{ 是映射.}$$

$$\text{若 } \varphi(xN_G(P)) = \varphi(yN_G(P)), \text{ 则 } xPx^{-1} = yPy^{-1}$$

$$\therefore x^{-1}yPy^{-1}x = P \quad \therefore (x^{-1}y)P(x^{-1}y)^{-1} = P$$

$$\therefore x^{-1}y \in N_G(P) \quad \therefore xN_G(P) = yN_G(P). \quad \therefore \varphi \text{ 是单射.}$$

$$\text{显然 } \varphi \text{ 是满射.} \quad \therefore \varphi \text{ 是双射.}$$

$$\alpha: G \times X \longrightarrow X \quad \text{是一个群作用 (验证省略).}$$

$$(g, xPx^{-1}) \mapsto (gx)P(gx)^{-1}$$

$$\begin{aligned} \varphi(\alpha(g, xN_G(P))) &= \varphi((gx)N_G(P)) = (gx)P(gx)^{-1} \\ &= gxPx^{-1}g^{-1} = \alpha(g, xPx^{-1}) = \alpha(g, \varphi(xN_G(P))) \end{aligned}$$

$$\therefore \varphi: G/N_G(P) \rightarrow X \text{ 是 } G\text{-集之间的同构.}$$

$$\therefore G \text{ 是有限群, } P \text{ 是 } G \text{ 的一个 Sylow } p\text{-子群}$$

$$\therefore P \triangleleft N_G(P) \quad \therefore P \leq N_G(P) \leq G$$

$$\therefore (G:P) = \underbrace{(G:N_G(P))}_{\in \mathbb{Z}_{p'}} \underbrace{(N_G(P):P)}_{\in \mathbb{Z}_{p'}}$$

$$\therefore \underbrace{|G|}_{\in \mathbb{Z}_{p'}} = \underbrace{|P|}_{\in \mathbb{Z}_{p'}} \underbrace{(G:P)}_{\in \mathbb{Z}_{p'}} \quad \therefore p \nmid (G:P) \quad \therefore p \nmid (G:N_G(P))$$

$$\because |X| = |G/N_G(P)| = (G:N_G(P)) \in \mathbb{Z}_{>1} \therefore p \nmid |X|$$

设 $H \leq G$ 且 H 是 p -群. 考虑群作用

$$\alpha: H \times X \longrightarrow X \quad (\text{显然这是群作用})$$

$$(g, xP_{X^{-1}}) \mapsto (gx)P(gx)^{-1}$$

$\because H$ 是 p -群, X 是非空有限集合

$$\therefore |X| \equiv |X^H| \pmod{p} \quad \because p \nmid |X| \quad \therefore p \nmid |X^H|$$

~~$$\therefore |X^H| \neq 0$$~~

$$\therefore |X^H| \neq 0 \quad \therefore |X^H| \in \mathbb{Z}_{>1}$$

$$\therefore \exists Q \in X^H \quad \therefore Q \in X^H \quad \therefore Q \in X \text{ 且 对 } \forall g \in H, \text{ 都有}$$

$$gQg^{-1} = Q \quad \therefore Q \in X \quad \therefore Q \leq G \text{ 且 } |Q| = |P|$$

$$\therefore H \leq N_G(Q)$$

$\because G$ 是有限群, p 是素数, Q 是 G 的 Sylow p -子群, H 是 G 的 p -子群, $H \leq N_G(Q) \quad \therefore H \leq Q$ 且 $H \leq Q$. ①得证

设 P_1, P_2 是 G 的任意两个 Sylow p -子群,

考虑群作用 $\alpha: P_1 \times X \longrightarrow X$

$$(g, xP_{X^{-1}}) \mapsto (gx)P(gx)^{-1}$$

$\because P_1$ 是 p -群, X 是非空有限集合.

$$\therefore |X| \equiv |X^{P_1}| \pmod{p} \quad \therefore p \nmid |X| \quad \therefore p \nmid |X^{P_1}|$$

$$\therefore |X^{P_1}| \neq 0. \quad \therefore |X^{P_1}| \in \mathbb{Z}_{\geq 1} \quad \therefore \exists Q_1 \in X^{P_1}$$

$$\therefore Q_1 \in X, \text{ 且对 } \forall g \in P_1, \text{ 都有 } gQ_1g^{-1} = Q_1$$

$$\therefore Q_1 \in X \quad \therefore Q_1 \text{ 是 } G \text{ 的 Sylow } p\text{-子群.}$$

$$\therefore P_1 \subseteq N_G(Q_1). \quad \therefore P_1 \subseteq Q_1 \quad \therefore |P_1| = |Q_1|$$

$$\therefore P_1 = Q_1 \in X. \text{ 同理可证 } P_2 \in X.$$

$$\therefore \exists g_1, g_2 \in G, \text{ s.t. } P_1 = g_1 P g_1^{-1}, \quad P_2 = g_2 P g_2^{-1}$$

$$\therefore g_2 g_1^{-1} \in G, \text{ 且 } (g_2 g_1^{-1}) P_1 (g_2 g_1^{-1})^{-1} = g_2 g_1^{-1} P_1 g_1 g_2^{-1}$$

$$= \cancel{g_2 g_1^{-1} g_1} P g_1^{-1} g_1 g_2^{-1} = g_2 P g_2^{-1} = P_2$$

$$\therefore P_1 \text{ 与 } P_2 \text{ 共轭.} \quad \textcircled{2} \text{ 得证.} \quad \square$$

定理 (Sylow 第三定理) G 是有限群, $|G| = p^m \cdot k$ ($m \in \mathbb{Z}_{\geq 1}$, $k \in \mathbb{Z}_{\geq 1}$, $p \nmid k$). 记 n_p 为 G 的 Sylow p -子群的个数, 则有:

$$\textcircled{1} n_p \mid k$$

$$\textcircled{2} n_p \equiv 1 \pmod{p}.$$

$$\textcircled{3} n_p = (G : N_G(P)), \quad P \text{ 是 } G \text{ 的一个任意的 Sylow } p\text{-子群.}$$

Proof: $\because G$ 是有限群, p 是素数 $\therefore G$ 有 Sylow p -子群.

设 P 是 G 的一个 Sylow p -子群. 定义集合 $X = \{gPg^{-1} \mid g \in G\}$.

则已经证明了集合 X 中的任意一个元素都是 G 的 Sylow p -子群.

已经证明了 $|X| = (G : N_G(P)) \in \mathbb{Z}_{\geq 1}$, 已经证明了 $p \nmid |X|$

设 H 是 G 的一个任意的 Sylow p -子群. 考虑群作用

$$\alpha: H \times X \rightarrow X$$

$$(g, xPx^{-1}) \mapsto (gx)P(gx)^{-1}$$

$\because H$ 是 p 群, X 是非空有限集合 $\therefore |X| \equiv |X^H| \pmod{p}$

$\because p \nmid |X| \therefore p \nmid |X^H| \therefore X^H \subseteq X$ 且 $p \nmid |X^H|$

$\therefore |X^H| \neq 0 \therefore |X^H| \in \mathbb{Z}_{\geq 1} \therefore \exists Q \in X^H$

$\therefore Q \in X$, 且对 $\forall g \in H$, 都有 $gQg^{-1} = Q$

$\therefore H \subseteq N_G(Q)$, Q 是 G 的 Sylow p -子群. $\Rightarrow$

$\because H \subseteq Q$ 且 $H \leq Q \therefore |H| = |Q| \therefore H = Q \in X$.

$\therefore X$ 是 G 的全体 Sylow p -子群组成的集合. $\therefore n_p = |X|$.

$\therefore n_p = |X| = (G : N_G(P)) \in \mathbb{Z}_{\geq 1}$

$\because G$ 是有限群, $P \leq N_G(P) \leq G \therefore (G : P) = (G : N_G(P))(N_G(P) : P)$

$$\therefore (G:P) = \frac{|G|}{|P|} = \frac{p^m \cdot k}{p^m} = k \quad \therefore k = n_p \cdot (N_G(P):P)$$

$$\therefore (N_G(P):P) \in \mathbb{Z}_{\geq 1} \quad \therefore n_p | k \quad \text{①得证.}$$

考虑 P 在 X 上的群作用: $\alpha: P \times X \longrightarrow X$
 $(g, xP_{x^{-1}}) \mapsto (gx)P(gx)^{-1}$

对 $\forall D \in X^P$, 有: $D \in X$, 且对 $\forall g \in P$, 有: $gDg^{-1} = D$

$$\therefore \cancel{P} \subseteq N_G(D) \quad \therefore P \subseteq D \text{ 且 } P \leq D.$$

$$\therefore |P| = |D| \quad \therefore P = D \quad \therefore D = P \quad \therefore X^P = \{P\}$$

$$\equiv \therefore |X| \equiv |X^P| \pmod{p} \quad \therefore n_p \equiv 1 \pmod{p} \quad \text{②得证}$$

设 B 是 G 的一个任意的 Sylow p -子群. 考虑群作用

$$\alpha: G \times X \longrightarrow X$$

$$(g, xP_{x^{-1}}) \mapsto (gx)P(gx)^{-1}$$

对于 X 中任意两个元素 $x_1P_{x_1^{-1}}, x_2P_{x_2^{-1}}$ (其中 $x_1, x_2 \in G$)

$$\text{有: } x_2x_1^{-1} \in G, \text{ 且 } (x_2x_1^{-1})(x_1P_{x_1^{-1}})(x_2x_1^{-1})^{-1}$$

$$= x_2P_{x_2^{-1}} \quad \therefore \text{这个群作用传递.}$$

$$\cancel{\text{Stab}_G} \therefore B \in X \quad \therefore \text{Stab}_G(B) = \{g \in G \mid gBg^{-1} = B\} = N_G(B)$$

$$\therefore |G/\text{Stab}_G(B)| = |GB| \quad \therefore |G/N_G(B)| = |X| = n_p$$

$$\therefore n_p = |G/N_G(B)| = (G : N_G(B)) \quad \textcircled{3} \text{得证.} \quad \square$$